

Fill-level sensors and connected garbage!

A Finnish start-up named Enevo Oy came up with an IoT-enabled waste-container monitoring service called Enevo One Collect. It uses low-power autonomous fill level and event sensors that use ultrasonic sensing and techniques to identify the status of the container it is monitoring. It also features temperature sensing, and comes in an IP67-class enclosure to endure extreme conditions. These sensors are wirelessly connected to a local network to send data.

The fill levels of waste bins and recycling containers are continuously monitored in real time using dedicated wireless sensors. The readings are sent to a server, which identifies the container location and analyses when and how the container is to be emptied. This information is sent to the waste management firm's logistic system. The One Collect method has saved 40% cost on waste collection and reduced environmental footprints.

For example, filtering is typically done in analogue domain using active filters. But the problem is that these filters are not tuneable individually and they consume high power. By transferring the signal to the digital domain and performing the filtering there, the filters become individually tuneable and can be switched off when not in use, which reduces power.

Harvesting ambient light. The integrated boost regulator ADP5090 by Analog Devices Inc. is one example where firms have come up with devices that can store energy from indoor ambient light and use the devices to power the sensors.

The ambient light is something that is freely and abundantly available in India throughout the year. Exploiting this natural resource as a viable power source has become a growing trend in wireless sensor network applications worldwide. It eliminates the need of power sources like batteries. In addition, it opens up huge possibilities of enabling IoT and connecting even remote/rural areas to the grid without the need for connected power or batteries.

Size and weight. No surprise that, just like lower power, size and weight of the gadget can also be an important consideration in the portable and wearable electronic space. There is a huge demand in market for smaller and lighter instruments. Size reduction is being achieved by merging circuit topology (whether the same block can be reused in multiple places) and different process technologies.

Integrating different sensors for higher scalability. Sometimes the same

application may require low-speed and high-speed sensing. Depending on the condition the application may switch back and forth between low-speed sensing and high-speed sensing.

The typical approach would be using two separate ADCs, one for low speed and the other for high speed, and keep one powered off when the other is being used.

However, when you use a device with wider range you can save space and money, the designing is easier, failure rate is reduced with fewer devices, and the power required is almost the same. Highly scalable devices also become useful in low-power applications. In IoT the power has to be locally derived through photocell or thermoelectric voltage generator.

The supply voltages generated by these are subjected to large variations. One need not worry about regulation in it, if they have an op-amp that can work over such large range; it avoids any intermediate stage of regulation.

Expandable platforms. It is very important to be able to incorporate new hardware and software solutions to a system in order to produce a better end product. As a result, it is very critical to have a flexible and expandable design platform that is compatible to and can incorporate both new hardware and software solutions, whenever necessary. Microchip's PIC16(L) F170X and PIC16(L)F171X family of 8-bit microcontrollers offers intelligent analogue and core-independent peripherals (configurable logic cell, complementary output generator and zero crossing detector), providing increased

flexibility for designers.

Somsubhro Pal Choudhury of Analog Devices believes that, the wide range of different types and sizes of data, bandwidth availability, reliability and cost considerations will not make a 'one size fits all' model. "Customisations and optimisations for each market vertical and sub-vertical would be the key with the industry severely fragmented, at least in this decade," he adds.

Dedicated processors. There has been a lot of talk about dedicated microcontrollers. Many original equipment manufacturers and fabrication plants are working on process technologies that could create processors optimised for and committed specifically to the analogue implementation in IoT applications.

In-built battery monitoring. As wearable devices are run by small batteries, it is necessary to optimise the use of these power sources. An accurate battery monitoring allows the user to know the exact state of charge of battery and adds intelligence to its usage.

For example, STC3115 battery monitoring IC by STMicroelectronics could give timely alert to the user for turning off the power-consuming blocks when battery is lower than the user-programmed threshold levels.

Innovative technologies driving cutting-edge analogue components

Shrinking the size and increasing performance has always been the challenge in semiconductor industry. The semiconductor industry has been witnessing the lowering of cost and size in the last couple of decades. Process nodes have been going from 45nm down to 28nm to 20nm, and will further go down to 16nm and 10nm in the next decade.

But the manufacturers' challenge was to achieve spatial, monetary or power efficiencies. Hence, as the limit of Moore's law reached 28 nm, they had to migrate from planar CMOS to a new process technology. That is when devices using FinFET technology were